C03 Tilemaps

CRE132 worksheet

What you'll make

A small outdoor area painted with **tiles**: grass, a dirt patch, a pond, flowers and stones, and a fenced pen. Instead of placing hundreds of sprites one by one, you paint them onto a grid like in a paint program.

You'll make a **Tile Palette**, paint on three **Tilemaps** (ground, decoration and foreground) that sort separately, and use a **Rule Tile**: one tile that picks the right fence piece by looking at its neighbours. Still no code.



Figure 1: The finished scene: grass with a dirt patch and a pond, flowers and stones, and a fenced pen with a gap. The player stands behind the top fence.

Before you start

1. Unzip `CRE132_C03_Tilemaps_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_C03_Tilemaps_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Core/C03_Tilemaps` and press **Play**. The Game view shows the player standing alone on a dark green background. Press Play again to stop.

The Hierarchy has a **Main Camera** (set up as in C02: **Size** 5.625 and a **Pixel Perfect Camera**), a **Global Light 2D** and the **Player**. You'll paint the world around the player.

1. The tile sheets

The tile art is in `Assets/_game/Art/Tilesets/MysticWoods`: `plains.png` (grass edges, dirt), `water1.png` (the pond), `decor_16x16.png` (flowers, stones, mushrooms) and `fences.png`. Click `plains.png` and look at its import settings. They're the C01 pixel-art settings plus two more:

- **Pixels Per Unit** 16, **Filter Mode Point (no filter)**, **Compression None**, as in C01.
- **Sprite Mode: Multiple**, and **Mesh Type: Full Rect**, so each tile fills its whole square.

Click **Open Sprite Editor**. The sheet has been cut with **Slice > Grid By Cell Size**, 16 × 16: every tile is one cell. Empty cells are skipped, so they don't become sprites. Close the Sprite Editor without changing anything.

2. A Tile Palette

A **Tile** is a small asset that points at a sprite. A **Tile Palette** is the box of tiles you paint with.

1. In the Project window, make a folder `Assets/_game/Tilemaps/MyTiles` for your tiles.
2. Open **Window > 2D > Tile Palette**. Dock the window next to the Inspector.
3. Click the palette drop-down at the top and choose **Create New Palette**. Call it `MyPalette`, click **Create** and save it in `Tilemaps/Palettes`.
4. Drag `plains.png` from the Project window into the palette window. When Unity asks for a folder, pick `Tilemaps/MyTiles`. Unity makes a Tile asset for every sprite in the sheet and lays them out in the palette.
5. Do the same with `water1.png`, `decor_16x16.png` and `Assets/_game/Art/Sprites/MysticWoods/grass.png`.

The project also has a ready-made palette, **MysticWoods**, in `Tilemaps/Palettes`, with its tiles in `Tilemaps/Tiles/MysticWoods`. Pick it in the palette drop-down to compare with yours. Using your own folder keeps your tiles separate from them.

3. A Grid and the Ground tilemap

Add it: choose **GameObject > 2D Object > Tilemap > Rectangular**. Unity makes a **Grid** with one **Tilemap** inside it. Rename the Tilemap **Ground**. On its **Tilemap Renderer**, set **Sorting Layer** to **Background** and **Order in Layer** to **0**.

- The **Grid** sets the cell size (1 × 1 unit, which at 16 pixels per unit fits a 16 × 16 tile).
- The **Tilemap** stores which tile is in each cell. The **Tilemap Renderer** draws them, and has a sorting layer like a Sprite Renderer.

4. Painting

In the Tile Palette window, check that **Active Tilemap** (just under the toolbar) says **Ground**. The tools you'll use:

Tool	Key	What it does
Brush	B	paints the selected tile(s)
Box Fill	U	fills a rectangle you drag out
Eraser	D	removes tiles
Picker	I	picks up tiles from the scene to paint with

1. **Grass:** click the grass tile in the palette, choose **Box Fill** and drag a rectangle that fills the camera's view (look at the Game view to check). The Finished one is 22 tiles wide and 14 high.
2. **Dirt patch:** in the palette, drag-select the whole 3 × 3 dirt block (edges, corners and centre) and paint it with the **Brush**. To make the patch bigger, paint more edge and centre tiles in between.
3. **Pond:** the same, with the 3 × 3 water block from `water1`.

Test it: the Scene view shows a grass field with a dirt patch and a pond. Use the Eraser to tidy any mistakes.

5. More tilemaps: Decoration and Foreground

Separate tilemaps let you sort things separately and, from C05, give some of them colliders.

Add it: right-click **Grid** in the Hierarchy and choose **2D Object > Tilemap > Rectangular**. Rename it **Decoration**. Do it again for **Foreground**. Set their Tilemap Renderers:

Tilemap	Sorting Layer	Order in Layer
Ground	Background	0
Decoration	Background	1
Foreground	Foreground	0

Set **Active Tilemap** to **Decoration** and paint flowers, stones, mushrooms and grass tufts from the `decor_16x16` tiles. They draw over the ground because they have a higher Order in Layer on the same layer.

Always check Active Tilemap before you paint. Painting onto the wrong tilemap is the most common mistake.

6. A Rule Tile for fences

A fence needs a different sprite for each piece: corners, straight bits, ends, a lone post. A **Rule Tile** picks the sprite by looking at its four neighbours, so you paint with one tile and the pieces sort themselves out.

Make one: in `Tilemaps/MyTiles`, right-click and choose **Create > 2D > Tiles > Rule Tile**. Name it `my_fence_rule`. In the Inspector:

1. Set **Default Sprite** to `fences_12` (the lone post, used when no rule matches).
2. Under **Tiling Rules**, click **+** to add a rule and set its sprite to `fences_14`. In the rule's 3 × 3 grid, click the left and right squares until they show a **green arrow** (a fence is there) and the top and bottom squares until they show a **red X** (no fence). This is the straight piece that runs left to right.
3. Add a rule for `fences_4`: green arrows up and down, red X left and right (the upright piece).
4. Add one for `fences_1`: green arrows down and right, red X up and left (the top-left corner).

That's enough to see how it works. A full fence needs 16 rules, one for each mix of neighbours. Open the finished one, `Tilemaps/Tiles/MysticWoods/fence_rule`, and look through its rules.

7. Painting with the Rule Tile

Drag `fence_rule` (the finished one) from the Project window into your palette. Set **Active Tilemap** to **Foreground** and paint a square pen with the **Brush**, leaving a gap in the bottom row. Paint the top row where the **Player** stands, so the fence crosses the player's feet. The Finished pen is 6 tiles wide and 6 high, with a gap of 2 tiles in the middle of the bottom row.

Test it: erase one piece in the middle of a side and watch its neighbours change into end pieces. Paint it back and they join up again.

8. The Foreground over the player

Select the **Player**: its Sprite Renderer is on the **Characters** sorting layer. If the fence doesn't cover its feet, move it with the **Move** tool (W) so it stands just behind the top fence. The fence draws over the player's legs because **Foreground** is below **Characters** in the sorting layer list.

Test it: now move the player in front of the bottom fence. The fence still draws over them, which looks wrong. That's expected: T03 (Y sorting) fixes it.

Check your work

- ☐ Your Grid has three Tilemaps: Ground, Decoration and Foreground, with the sorting layers in the table above.
- ☐ Grass covers the view, with a dirt patch and a pond with proper edges.
- ☐ Flowers and stones draw on top of the ground.
- ☐ The fence pen joins up at the corners, and erasing a piece updates its neighbours.
- ☐ The fence draws over the player standing behind it.
- ☐ Compare with `CRE132_C03_Tilemaps_Finished` if something looks different.

Key points

- **A Tile is an asset that points at a sprite.** The Tilemap stores which tile goes in each cell, and the Tilemap Renderer draws them.
- **Use separate Tilemaps** for things you want to sort or edit separately (and, in C05, give colliders to).
- **A Rule Tile picks its sprite from its neighbours**, so you paint with one tile and the edges sort themselves out.
- **The Foreground always draws over the player**, even in front of the fence. T03 fixes that with Y sorting.

If something goes wrong

- **I painted on the wrong tilemap:** check **Active Tilemap** in the Tile Palette window. Erase, switch, and paint again.
- **My tiles are invisible:** the tilemap is under the grass. Check its Tilemap Renderer's Sorting Layer and Order in Layer.
- **Tiles have thin gaps or blurry edges:** the sheet's Filter Mode isn't **Point**, or its Mesh Type isn't **Full Rect**.
- **The palette is empty or greyed out:** pick your palette in the drop-down at the top of the Tile Palette window.
- **My Rule Tile only paints lone posts:** no rule matches. Check the arrows and Xs in each rule, and remember a fence needs 16 rules to cover every case.

Try it

1. Paint a second pen with a gate, or a single line of fence.
2. Make a Rule Tile for the pond using the 3 × 3 water block. Corners and edges also need the diagonal neighbours.
3. Look up the **Animated Tile** in the 2D Tilemap Extras package. How could you make the pond ripple? (The full Mystic Woods pack has six water frames; this project only has the first.)